

Khalil LAGHA

M2 student - Electrical Energy Systems Design (CSEE) · Université Grenoble Alpes

Ranked 2nd of the M1 EEA cohort (15.43/20). DC-DC and AC-DC converters, PI/PID loop tuning and EMI compliance, validated in simulation then on real hardware across five internships, from the lab (GIPSA-lab) to industrial sites.

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- Authorized to work in France



01 Experience

- Research Intern · GIPSA-lab 04/2026 - 07/2026 · 3.5 months
 - Modeled a **10-cell PEMFC fuel-cell stack** in Matlab/Simulink: **6%** dynamic and **1%** static error against real measurements.
 - Tuned a **DC-DC boost converter PI regulation loop** to optimize efficiency; validated first in simulation, then on a real test bench.
- Intern · SLB (Schlumberger) 07/2025 · 1 month
 - Authored technical syntheses **in English** (directional drilling) in an international industrial environment.
- Power Electronics & Automation 01/2024 · 15 d + 09/2024 · 1 month Intern · EURL Lagha
 - Contributed to an **industrial DC-DC buck converter**: 50 V input, adjustable 0-48 V output, **25 A**.
 - Contributed to **4 transformer-rectifiers** feeding pipelines (electrolysis): three-phase 400 V → 100 V transformer, rectification + capacitive filtering, **100 V DC / 80 A (≈ 8 kW)** output, ripple < 5%, EMI compliance.
 - Co-designed the supervision HMI (Siemens TIA Portal, Ladder PLC); analyzed a customer specification, sized the corresponding transformer and produced the electrical schematics in **EPLAN**.
- Power Grid Intern · Sonelgaz 03/2024 · 15 d · observation
 - Analyzed the full generation → distribution chain (**EHV 400/220 kV → HV 60 kV → MV 30/10 kV → LV 400/230 V**) and substation architecture; surveyed an MV/LV substation during a technical site visit.

02 Projects

Power grid analysis · Matlab · solo
Load flows from **3 to 118 buses** (PQ/PV/slack; IEEE 14/30/57/118 networks); benchmarked **3 numerical methods**: Gauss-Seidel, Newton-Raphson, fast-decoupled (accuracy vs computation time); studied **transient stability**.

Field-oriented control (FOC) of a **5 kW / 380 V induction motor** · Python + Simulink
Indirect rotor-flux orientation; **three matching implementations**, incl. Simscape Electrical at 10 kHz switching. Open source.

Electrical machine sizing · Python (OOP, PyQt) · team
Stator winding design, flux density via reluctance networks, finite-element validation (FEMM), automatic parameter generation.

Neural network from scratch (MLP) · Matlab · open source (MIT)
Multilayer perceptron and backpropagation hand-coded, no toolbox; debugged, tested and published: github.com/KhalilEllahLAGHA/mlp-backpropagation-matlab.

Also: cascade control of a DC motor (three-phase thyristor bridge, Simulink) · Siemens Step7 automation (HMI + Ladder) · Arduino line-follower robot (Poly Maze competition).

2nd

of the M1 EEA cohort (UGA)

15.43

M1 annual average /20

C2

English

03 Education

M2 CSEE - Electrical Energy Systems Design from 09/2026
Université Grenoble Alpes · work-study

M1 EEA (EECS) 2025 – 2026
Université Grenoble Alpes · 2nd of the cohort, annual average 15.43/20

Engineering cycle, Electrical Engineering 2021 – 2025
École Nationale Polytechnique, Algiers · 4 of 5 years completed; prep: 16th/358 then 49th/264

Baccalauréat in Mathematics 2021
Highest honours (16.93/20)

04 Skills

Power electronics

DC-DC / AC-DC conversion

Machines & transformers

Grids: load flow, stability

PID/LQR loops · Kalman · FOC

PLC / HMI / SCADA

Electrical CAD: EPLAN, AutoCAD

AI: NN, GA, fuzzy logic

05 Software

Matlab/Simulink (Simscape Power Systems) · Python (OOP, PyQt) · C/C++ · LTspice · Proteus · EPLAN · AutoCAD Electrical · Siemens TIA Portal · Step7 (Ladder/SCL) · FEMM · SolidWorks · Arduino · Git/GitHub

06 Languages & certification

French C1
English C2
SLB NEST & SIPP Level 2 · 07/2025 (HSE)

07 Soft skills

Fast learner

Rigorous

Organized

Problem solving

Comfortable with tools